

Jiwon Heo

 [Homepage](#)

 4865sj@gmail.com

 [linkedin.com/in/4865sj/](https://www.linkedin.com/in/4865sj/)

 github.com/4865sj

Education

KAIST

Ph.D.

Korea University

Bachelor

- **Physics:** 1st Major, GPA 4.03/4.50
- **Mathematics:** Double Major, GPA 4.21/4.50
- **Economics:** Minor, GPA 4.08/4.50
- Total GPA 4.07/4.50

September 2024 - Now

Daejeon, Republic of Korea

March 2019 - August 2024

Seoul, Republic of Korea

Experience

Signal Processing & Advanced Intelligence (SPAI) LAB (Prof. Jiwon Yoon)

Intern

July 2022 – December 2023

Seoul, Republic of Korea

Quantum information theory (QIT) LAB (Prof. Changhun Oh)

Ph.D.

September 2024 - Now

Daejeon, Republic of Korea

Paper list

Classical simulation of a quantum circuit with noisy magic inputs

- Proposed a novel classical simulation algorithm for simulating a quantum circuit with noisy magic inputs.
- Our algorithm is established in a general setting; hence, we apply our algorithm to several scenarios and analyze them.

Accelerating secure permutation: Application to matrix algebra

- Proposed a new, rapid method for multiplying ciphertext by a specialized form of plaintext, leveraging the algebraic properties of ciphertext rather than traditional homomorphic multiplication
- Exhibited theoretical and experimental improvements and availability by applying it to matrix multiplication.

Speed-up of Data Analysis with Kernel Trick in Encrypted Domain

Fast private location-based information retrieval over the torus

Patent list

캐노니컬 임베딩을 사용하는 원분다항식몫체 기반의 암호문에서 특정 원소를 빠르게 추출하는 장치 및 방법

Domestic patent (Application no. 10-2023-0124483) (Proceeding)

- Proposed a new efficient operation about extracting specific elements from ciphertext based on cyclotomic polynomial quotient field, utilizing its algebraic properties.
- Led proactively patent application process, including drafting transmittal letter and meeting with patent attorney.

Core projects

Numerical analysis for ODE | Python

PHYS233

- Conducted numerical analysis of ordinary differential equations corresponding to classical dynamics with friction.

Hypothesis testing based on ML and its applications to physics experiments | Python, Tensorflow

PHYS466

- Designed and implemented hypothesis testing based on machine learning and apply to real experimental data.

Implementation of hypothesis testing | Python

PHYS303

- Embodied t-test and applied it to real data in order to determine whether experimental data can explain its corresponding theory.

Implementation of Bayesian statistics | Python

STAT404

- Implemented basic concepts of Bayesian statistics and applied them to real data.